

Rubric Exam ADA (Problem 1: Electronic Circuit Optimization)

Question 1 — a) Data structures (2 points)

Describe the data structures needed to model this problem. Define the relevant variables and provide a graphical representation of these structures.

Criterion	Points
Recognizes that a graph must be used	0.5
Recognizes structures (adjacency matrix or adjacency set) as suitable for implementing it	0.4
Recognizes that the edges must be weighted	0.4
Recognizes vertices as pins and edges as cables (even if not stated explicitly)	0.3
Provides a graphical representation of the structures	0.4

Question 2 — b) Optimization algorithms (1 point)

State known algorithms that solve this optimization problem. Briefly describe each algorithm and justify its suitability for meeting the company's requirements.

Criterion	Points
Mentions Prim and Kruskal as suitable algorithms	0.3
Does not mention unrelated algorithms (e.g. DFS/BFS)	0.1
Correctly defines Prim (random starting vertex + lowest-weight adjacent edge)	0.2
Correctly defines Kruskal (adds lowest-weight edges globally)	0.2
Recognizes that neither algorithm can form cycles	0.1
Recognizes the complexity of both algorithms	0.1

Question 3 — d) Algorithm implementation (3 points)

Implement an algorithm that, given the optimized circuit resulting from the previous section, identifies the two pins whose distance, defined as the length of the shortest path between them, is maximum. The algorithm must return this distance and the intermediate pins that make up the shortest path between the corresponding pair of pins. The code must include comments explaining how it works. Show an example call to the algorithm from a main program, specifying the initial parameter values.

Criterion	Points
Solves it with an optimal approach based on BFS or DFS	0.6
Solves it with a non-optimal approach based on Floyd-Warshall	0.4
If using Floyd-Warshall, implements the Path function to obtain intermediate vertices (not just mentions it)	0.5

Comments the code sufficiently well	0.5
The code is free of errors	1.0

Total Problem 1: 6.0

Note: section c) of the original exam is excluded from this rubric due to PDF scanning issues; it was not used in the study.